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Acute steroid-refractory GI GVHD is not associated with significant differences in gut taxonomic composition compared to steroid-sensitive GI GVHD immediately prior to onset of disease

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Abstract

Background: Taxonomic composition of the gut microbiota at the time of neutrophil engraftment is associated with the development of acute gastrointestinal graft-versus-host disease (GI GVHD) in patients undergoing allogeneic hematopoietic stem cell transplantation. However,

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less is known about the relationship between the gut microbiota and development of steroid-refractory GI GVHD immediately prior to the onset of disease.

Objectives: Markers of steroid-refractory GI GVHD are needed in order to identify patients who may benefit from the early initiation of non-corticosteroid-based GVHD treatment. Our aim was to identify differences in taxonomic composition in stool samples from patients without GVHD, with steroid-responsive GVHD and with steroid-refractory GI GVHD in order to identify predictive microbiome biomarkers of steroid-refractory GI GVHD.

Study design: We conducted a retrospective case-control, single institution study, performing shotgun metagenomic sequencing on stool samples from patients with (n=36) and without GVHD (n=34) matched for time since transplant. We compared the taxonomic composition of the gut microbiome in those with steroid-sensitive GI GVHD (n=17) and steroid-refractory GI GVHD (n=19) to each other and to those without GVHD. We also performed associations between steroid-refractory GI GVHD, gut taxonomic composition, and fecal calprotectin, a marker of GI GVHD to develop composite fecal markers of steroid-refractory GVHD prior to the onset of GI disease.

Results: We found that fecal samples within 30 days of GVHD onset from patients with and without GVHD or with and without steroid-refractory GI GVHD did not differ significantly in Shannon diversity (alpha-diversity) or in overall taxonomic composition (beta-diversity). While those patients without GVHD had higher relative abundance of *Clostridium* spp., those with and without steroid-refractory GI GVHD did not significantly differ in taxonomic composition between one another. In our study, fecal calprotectin prior to disease onset was significantly higher in patients with GVHD compared to those without GVHD and higher in patients with steroid-refractory GI GVHD compared to steroid-sensitive GI GVHD. No taxa were significantly associated with higher levels of calprotectin.

INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (HCT) is a potentially curative strategy for patients with a variety of hematological malignancies. Widespread success of HCT has been hindered by life-threatening post-transplant complications, including relapse, and immunological or infectious sequelae. Acute graft-versus host disease (GVHD), characterized by the activation of donor-derived alloreactive T cells and an imbalance in effector and regulatory arms of the immune system^{1,2}, is a major source of mortality and morbidity. Acute GVHD potentiates a pro-inflammatory positive feedback loop that can cause profound damage to host tissues, namely the gut, skin, and liver³. Notoriously the most difficult manifestation to treat, involvement of the gastrointestinal (GI) tract by GVHD commonly presents as severe abdominal cramping, nausea, vomiting, and profuse diarrhea⁴. High-dose systemic corticosteroids are the current first-line therapy, however nearly half of patients do not respond⁵. While ruxolitinib has now been approved for second-line treatment of acute GVHD, those with steroid-refractory GVHD continue to suffer mortality rates in excess of 70–90%⁶.

Early detection and treatment escalation are critical in improving outcomes for steroid-refractory GVHD. Identifying specific biomarkers to predict responsiveness to

immunosuppressive agents is therefore of substantial clinical relevance. Measurement of serum markers demonstrate improved prognostic value relative to clinical assessments. Suppressor of tumorigenicity-2 (ST2) and regenerating islet-derived protein 3-alpha (REG3a) have been previously shown to predict resistance to steroids⁷. Fecal markers, including calprotectin, albumin, and alpha-1-antitrypsin, have gained attention for their efficacy in screening for inflammatory bowel disease and other inflammatory conditions⁸. Whether they can be utilized the setting of acute GVHD and steroid sensitivity is equivocal, however, and more robust validation is needed. To date, no biomarkers for steroid-refractory GVHD have been widely implemented in clinical practice.

There is emerging interest in developing gut microbiota-based markers for evaluating susceptibility to steroid-refractory GVHD. The gut microbiota, referring to the trillions of microorganisms that reside in the intestine, holds essential functions in immune system development and modulation, among other aspects of host physiology. Alterations in gut microbial composition have been associated with acute GVHD and GVHD-related mortality^{5,9–13}. Decreased diversity of the microbiota, a classic marker of microbial dysbiosis, correlates not only with an increased incidence of acute GVHD, but also with infectious complications, and overall mortality^{5,9–11,13–15}. Additionally, an enrichment of anaerobic genera, such as *Blautia* and *Clostridiales*, accompanies improved survivorship and decreased GVHD incidence^{9,10}. Few studies have investigated whether specific microbial signatures can predict steroid resistance. *Golob et al.* demonstrated that steroid-refractory GVHD is strongly associated with the presence of butyrate-producing bacteria prior to disease onset⁵. Another study of 9 patients found a ratio of Dorea to Akkermansia gut relative abundance may be associated with steroid responsiveness¹⁶. There is also non-randomized evidence of improved symptoms and clinical outcome in steroid-refractory GVHD patients receiving fecal microbial transplantation, further suggesting a link between microbiota dysbiosis and steroid-refractory GVHD pathophysiology^{17–19}. Altogether, current challenges in early steroid-refractory GVHD detection argue for an extension of biomarker discovery efforts to the gut microbiota.

Towards this end, we used a high-throughput metagenomic sequencing approach to describe putative taxonomic and functional characteristics of steroid-refractory GVHD in the gut microbiota in a pilot, retrospective cohort of GVHD patients. Additionally, we relate these findings to the extent of intestinal inflammation as measured by fecal calprotectin and compare the ability of these markers to predict steroid-refractory GVHD. We hypothesized that, similar to prior studies^{9–13}, patients with GVHD would have lower gut microbiota diversity immediately prior to disease onset. Additionally, we reasoned that patients with steroid-refractory GVHD would be at the extreme lower end of diversity and that just as calprotectin could serve as a marker of disease, gut microbial taxonomic composition would parallel calprotectin in predicting steroid-refractory GI GVHD.

METHODS

Cohort Selection

Stool samples were obtained retrospectively from a stool biospecimen repository of patients undergoing hematopoietic stem cell transplantation (HCT) at Stanford Hospital between

2015 and 2019. Research activities were supported the Institutional Review Boards of Stanford University under IRB protocol 42545 (PI: Ami Bhatt) with consent obtained for all research subjects. Patients were selected for the study cohort if they (1) had undergone allogeneic stem cell transplant without restriction on conditioning regimen, type of transplant, or underlying malignancy, (2) underwent colon biopsy for suspected acute gastrointestinal GVHD, (3) had one stool specimen collected within 30 days prior or up to 3 days following biopsy, and (4) were exposed to fewer than two days of corticosteroids leading up to stool specimen collection. Control subjects were defined as patients who received allogeneic stem cell transplant, but never developed acute GVHD of the lower gastrointestinal tract or otherwise. Stool specimens from control samples were matched with GVHD samples according to the interval between the HCT date and date of stool collection (+/- five days). Steroid-refractory GVHD was defined by the clinical team as disease progression after 5 days of steroid therapy or no response to steroid treatment by day 7. While treatment for steroid-refractory GVHD varied between patients, all patients with an initial diagnosis of GVHD of grade II or more received the equivalent of prednisone 1mg/kg once or twice daily.

Metagenomic sequencing of stool samples

Stools in the biorepository had previously been collected on a weekly basis fresh from HCT patients hospitalized at Stanford, stored at 4°C for less than 24 hours before aliquoting into cryovials, and frozen at -80°C prior to DNA extraction. DNA was extracted from stool samples using the QIAamp DNA Stool Mini Kit (QIAGEN) according to the manufacturer's instructions, including an additional bead-beating step prior to extraction with the Mini-Beadbeater-16 (Biospec Products) and 1 mm diameter Zirconia/Silica beads (BioSpec Products). Bead-beating consisted of 7 rounds of alternating 30 second bead-beating bursts followed by 30 seconds cooling on ice. Qubit Fluorometric Quantitation (Life Technologies) was used for measurement of DNA concentration for all samples. DNA sequencing libraries were prepared with the Nextera XT DNA Library Prep Kit (Illumina) and library concentrations were measured with Qubit Fluorometric Quantitation. If DNA library yield was insufficient for downstream sequencing, library preparation was repeated with the Nextera Flex Library Prep Kit (Illumina). Prepared libraries were multiplexed and sequenced on the HiSeq 4000 platform (Illumina) to generate 100 base pair paired-end reads.

Calprotectin Quantification

Fecal calprotectin concentrations were assessed with the QUANTA Lite® Calprotectin Extended Range Enzyme-linked immunoabsorbent assay (ELISA) (Inova), in accordance with manufacturer's instructions. Samples were tested in triplicate and applied randomly to antibody-coated plate to minimize batch and edge effects. OD values were measured at 450 nm. We imputed final calprotectin concentrations with a logit-log 4 parameter regression, based on OD readings of supplied standard solutions.

Computational and Statistical Methods

To remove PCR artifacts and optical duplicates, stool metagenomic sequencing reads were deduplicated using seqtk rmdup v1.3-r106 with default parameters²⁰. Deduplicated reads were then trimmed with TrimGalore v0.5.0²¹, a wrapper for CutAdapt v1.18²² specifying

a minimum quality score of 30 for trimming (--q 30) and minimum read length of 60 (--length 60). The Burrows Wheeler Aligner (BWA)v0.7.17-r1188 was applied for removal of reads aligning to the human genome (hg19)²³. The Kraken v2.0.7-beta taxonomic classifier with default parameters enabled characterization of the microbial content of the processed metagenomic sequencing samples²⁴. We curated a comprehensive custom reference database containing all bacterial and archaeal genomes in Genbank assembled to “complete genome,” “chromosome” or “scaffold” quality as of February 2019. Species abundance at each taxonomic rank was re-estimated with Bracken v2.0.0²⁵. Unmapped and unintegrated pathways were excluded for further analysis. Pathway abundance from the Humann3 pipeline and raw counts from the Bracken pipeline were normalized using the formula:

$$\log_{10}\left(\frac{RC}{n} \times \frac{\sum x}{N} + 1\right)$$

where RC represents the individual sample OTU counts, n is the number of sequences in a sample, the sum of x is the total number of counts in the table and N is the total number of samples. To remove potentially spurious low abundance taxa, we removed all taxa or pathways with a relative abundance below 0.000688% at phylum level, 0.0286% at genus level, 0.0286% at species level, and 0.00235% of potential metabolic pathways, based on histogram distribution (data not shown) and corresponding to a mean relative abundance of 100 reads or more. Pairwise analysis was performed between (1) the control subjects and GVHD patients, and (2) between the steroid-sensitive and steroid-refractory GVHD patients among the pathways and relative abundance using t-tests. Normalized and filtered relative abundance were compared at phylum, genus, and species level.

Principal coordinate analysis (PCoA) was performed with R package “vegan” (version 2.5.7) at each taxonomic level with the “Bray-Curtis” distance in the “capscale” function, and the p-values were extracted from the PERMANOVA tests using the “adonis” function. MDS plots were produced using the “ordiplot” and “ordiellipse” functions with confidence limit of 0.95, where the ellipses represent the standard error of the weighted centroids. Shannon diversity was calculated using the “diversity” function from the “vegan” package, and p-values were extracted from the Wilcoxon test using the “wilcox_test” function from the “coin” package (version 1.4.2). Differential abundance analysis was performed using the “t.test” function, “wilcox.test” function, and ALDEx2 package (version 1.30.0) with default settings in R studio. Calprotectin levels were compared using the wilcox_test” function from the “coin” package. Kendall correlation p-values between calprotectin levels and samples groups were calculated through base-R “cor.test” function. All statistical analyses were performed using R (version 4.1.2). A p-value of 0.05 was defined as the threshold for statistical significance, and, where applicable, a Benjamini-Hochberg correction was applied for multiple comparisons.

RESULTS

In the construction of our matched case-control study (see methods), we identified 38 patients with acute GI GVHD who met inclusion criteria and of those with GVHD, 19

were found with steroid-responsive and 19 with steroid-refractory acute GI GVHD. Stools from HCT patients with acute GI GVHD were found in our biorepository; those that met the following specific inclusion criteria were included in this study: patients had had one stool specimen collected within 30 days prior or up to 3 days following GVHD diagnosis and were exposed to fewer than two days of corticosteroids leading up to stool specimen collection (for full criteria see methods). Control subjects (n=38) were defined as HCT patients who never developed any form of acute GVHD but whose samples were matched to those with GVHD based on timing of stool collection relative to transplant (Supplementary Figure S1).

No differences were found between patients within each of 3 categories (no GVHD, steroid-refractory GVHD, and steroid-sensitive GVHD) in demographic characteristics, underlying malignancy, or transplant characteristics (Table 1). While steroid-sensitive and no GVHD groups were similar in 1-year overall survival (78.9%,) those with steroid-refractory GVHD had less than a 30% overall survival after 1 year. Groups did not differ in *Clostridioides difficile* colitis incidence or antibiotic use within the 30 days prior to stool collection.

Gut microbial taxonomic composition does not differ between groups based on acute GI GVHD

Stool samples from each patient were subjected to shotgun metagenomic sequencing to a target depth of 30–50 million 100 base pair paired-end reads. Four samples (2 steroid-sensitive, 2 no-GVHD) failed library preparation, and 2 control samples initially identified were outside of the required matching window, resulting in a total of 34 samples from patients without GVHD and 36 from those with acute GI GVHD (17 steroid-sensitive, 19 steroid-refractory). Sequenced reads were filtered to retain high-quality non-duplicated reads and human sequences removed. Reads were aligned into contigs and taxonomically categorized with gene prediction performed. Samples yielded a mean of 5,932,711 reads (range 6,454 to 14,744,046) at the species level after quality filtration and taxonomic assignment.

In order to compare taxonomic composition between patients with and without GVHD or with steroid-sensitive versus steroid-refractory GVHD, we calculated Shannon diversity at each taxonomic level but did not find any significant differences or consistent trends (Figure 1 at the genus level; Supplementary Figure S2 at phylum and species levels). Differences in Shannon diversity of predicted pathways were observed between groups patients with and without GVHD, but not between steroid-sensitive versus steroid-refractory GVHD groups (Supplementary Figure S3). We compared Bray-Curtis dissimilarity indices between groups in multidimensional scaling plots and again show no clear separation between the GI GVHD and non-GVHD groups or between the steroid-sensitive and steroid-refractory groups for taxa (Figure 2) and metabolic pathways (Supplementary Figure S4).

To provide a more detailed determination of potential microbial compositional differences, we performed differential abundance analysis using t-tests to compare groups (GI GVHD vs. non-GVHD and steroid-refractory vs. steroid-sensitive GI GVHD, adjusted for multiple testing). We found differences in relative abundance at the species levels including higher *Clostridium* spp. in patients without GVHD compared to those with GI GVHD

(Supplementary Table 1 and Supplementary Figure S5). However, we did not see any differences between steroid-sensitive and steroid-refractory groups after FDR correction (Supplementary Table 1 and Supplementary Figure S6). No differences in metabolic pathways were observed between any groups (Supplementary Table 1 and Supplementary Figures S7 and S8).

Calprotectin level

Calprotectin has demonstrated efficacy in differentiating patients with steroid-refractory and steroid-sensitive GVHD^{4,8}. As expected, we found calprotectin levels to be significantly higher in patients with GVHD compared to no GVHD, and in patients with steroid-refractory GVHD compared to steroid-sensitive GVHD (Figure 3). We next aimed to therefore determine whether specific taxa were associated with higher calprotectin levels and found that although several taxa were correlated with higher calprotectin levels, none were significant after correction for multiple hypothesis testing (Supplementary Table 2, Supplementary Figure S9).

DISCUSSION

The gut microbiota is an increasingly well-characterized marker of outcomes following HCT. Prior studies have demonstrated that the diversity of the gut microbiota prior to transplant and at the time of engraftment is correlated with acute GVHD and GVHD-related mortality as well as overall survival. Little has been written about the state of the gut microbiota immediately preceding the onset of GVHD and as a marker to predict response to GVHD treatment. We performed a retrospective matched case-control study to determine taxonomic associations with steroid-refractory GVHD and a marker of steroid-refractory disease, calprotectin. While we confirmed many of the same associations seen in prior studies (e.g., elevated *Clostridium* spp. in patients without GVHD)^{9–13} we were unable to find any differences in alpha or beta-diversity between steroid-sensitive and refractory disease, nor were we able to identify any associated taxa with refractory disease or fecal calprotectin levels.

While this study represents the largest of its kind, most likely we were limited by the number of patients included, the retrospective nature of the study increasing the risk for unrecognized bias, and potentially by the timing of stool sample collection. Prior studies illustrating taxonomic markers of steroid-refractory disease have used samples obtained at the time of engraftment, prior to transplant, or more than 30 days from GVHD onset^{5,16}. There may not in fact be a gut microbial marker of steroid response proximal to the onset of gut GVHD symptoms. Publishing negative results of a microbiota study in the HCT population is a valuable contribution to the research in impacting subsequent research and study design. Additionally, the post-transplant immune repertoire following transplant may have already been established by this time, and the subsequent cascade of alloreactivity set in motion. Alternatively, a very large sample size may enable differentiation of signal from noise at this timepoint. For example, *Citrobacter* was the most different taxon between groups and was higher in relative abundance in those with steroid refractory versus steroid sensitive. We ran simulations with *Citrobacter* using the demonstrated effect size of 0.21.

To achieve 80% power after FDR correction at a corrected p-value of 0.05, on average samples from 600 study participants would be required (300 for steroid-sensitive and 300 for steroid-refractory GI GVHD).

In our bioinformatic pipeline we used the data from Bracken outputs and compared these same outputs to MetaPhlAn3²⁶ (data not shown). We found that while Bracken was able to identify a greater number and diversity of taxa, MetaPhlAn3 characterized taxa more conservatively and did not output the same diversity of organisms within stool samples. This has been demonstrated previously in the literature²⁷. We therefore chose to use Bracken-based taxonomic classification with a stringent threshold aimed to filter out taxa and pathways with mean abundance less than 100 reads after normalization among the samples. While this removed much of the potential contamination and spurious “phantom” classifications that Kraken and Bracken are prone to, we may have discarded some important low abundance taxa relevant to differences in outcomes for patients with steroid-refractory versus steroid-sensitive GVHD. As in much of the microbiota literature in this heterogeneous population of patients, the ratio of signal to noise may not allow for the differentiation of potentially important low abundance taxa that could otherwise serve as markers for disease.

The prediction of steroid responsiveness in patients with GVHD remains a valuable and clinically relevant goal. Given that the gut microbiota has been implicated in the development of acute GVHD, further studies to determine the timing and taxonomic changes that are the most predictive are will be necessary to attain this goal. Ultimately, a personalized approach to the prevention, diagnosis, and treatment of GVHD is likely necessary, and will require longitudinal sampling from a large multi-center cohort of patients to understand how to mitigate the most devastating complications of allogeneic HCT in the future.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- In a small cohort of 70 patients, fecal samples within 30 days of acute GI GVHD onset from patients with and without GVHD or with and without steroid-refractory GI GVHD did not differ significantly in Shannon diversity (alpha-diversity) or in overall taxonomic composition (beta-diversity).
- Fecal calprotectin prior to disease onset was significantly higher in patients with GVHD compared to those without GVHD and higher in patients with steroid-refractory GI GVHD compared to steroid-sensitive GI GVHD.
- Patients without acute GVHD had higher gut relative abundance of *Clostridium* spp.
- Those with and without steroid-refractory GI GVHD did not significantly differ in taxonomic composition between one another.
- No taxa were significantly associated with higher levels of fecal calprotectin.

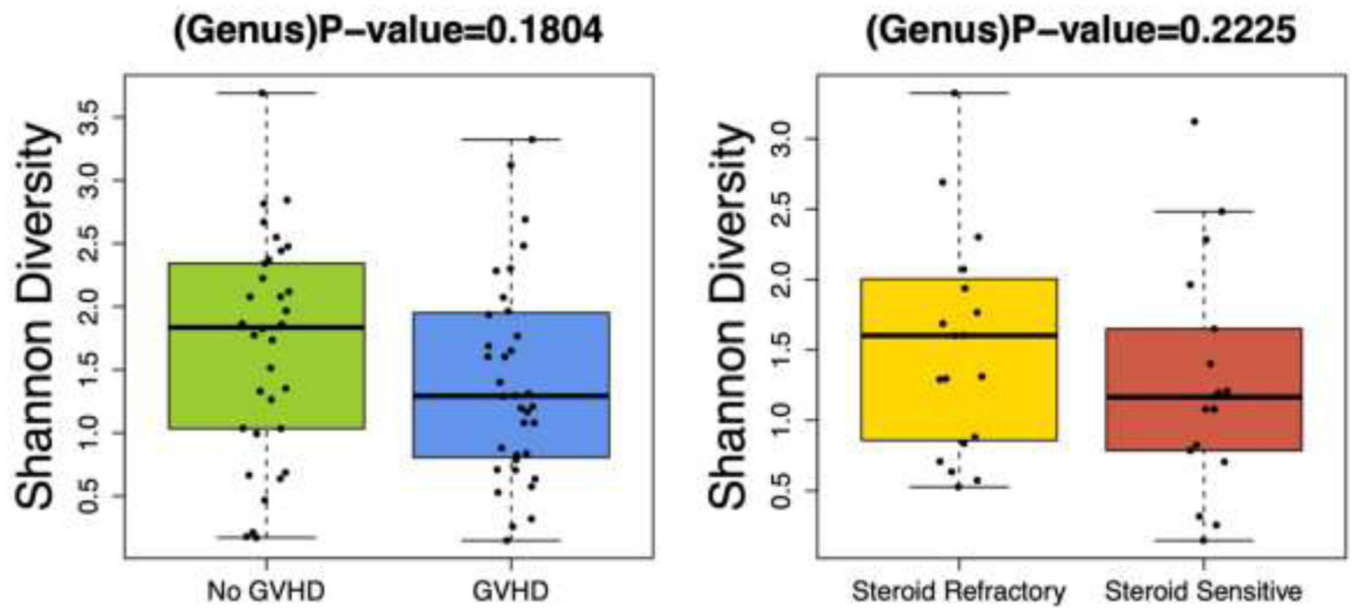


Figure 1. Differences in Shannon diversity at the genus level were not observed between patient groups.

Shannon diversity was compared between patients with or without GI GVHD and between patients with steroid-refractory versus steroid-sensitive GI GVHD using the Wilcoxon rank sum test. Outliers (1 in the GVHD vs. No GVHD plot, 1 in the Steroid Refractory vs. Steroid Sensitive plot) were included in the statistical models and excluded from bar plots to improve visualization.

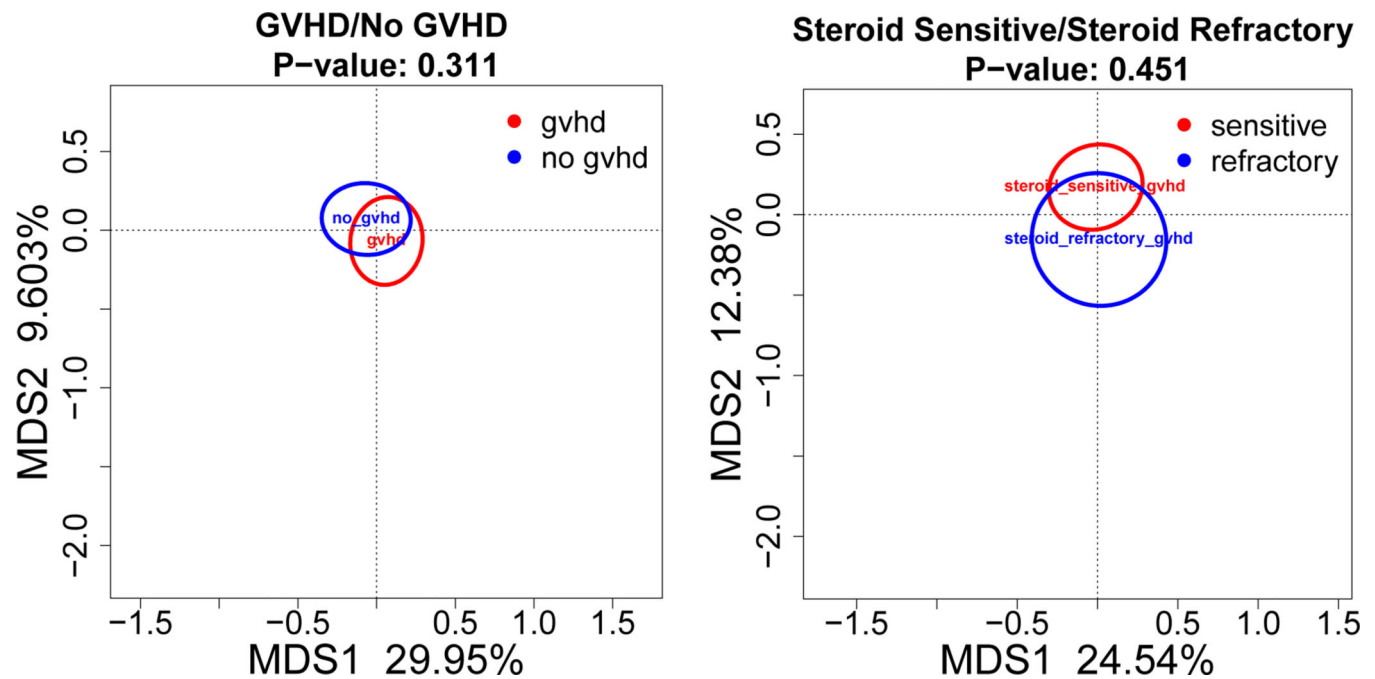


Figure 2. Gut microbial taxonomic composition prior to GI GVHD onset is not significantly different at the genus level between patient groups.

Multidimensional scaling (MDS) of gut microbiome taxonomic composition at the genus level was performed using Bray-Curtis dissimilarity. The Adonis test was used to compare gut microbial taxonomic composition between patients with and without GI GVHD and with steroid-refractory versus steroid-sensitive GI GVHD.

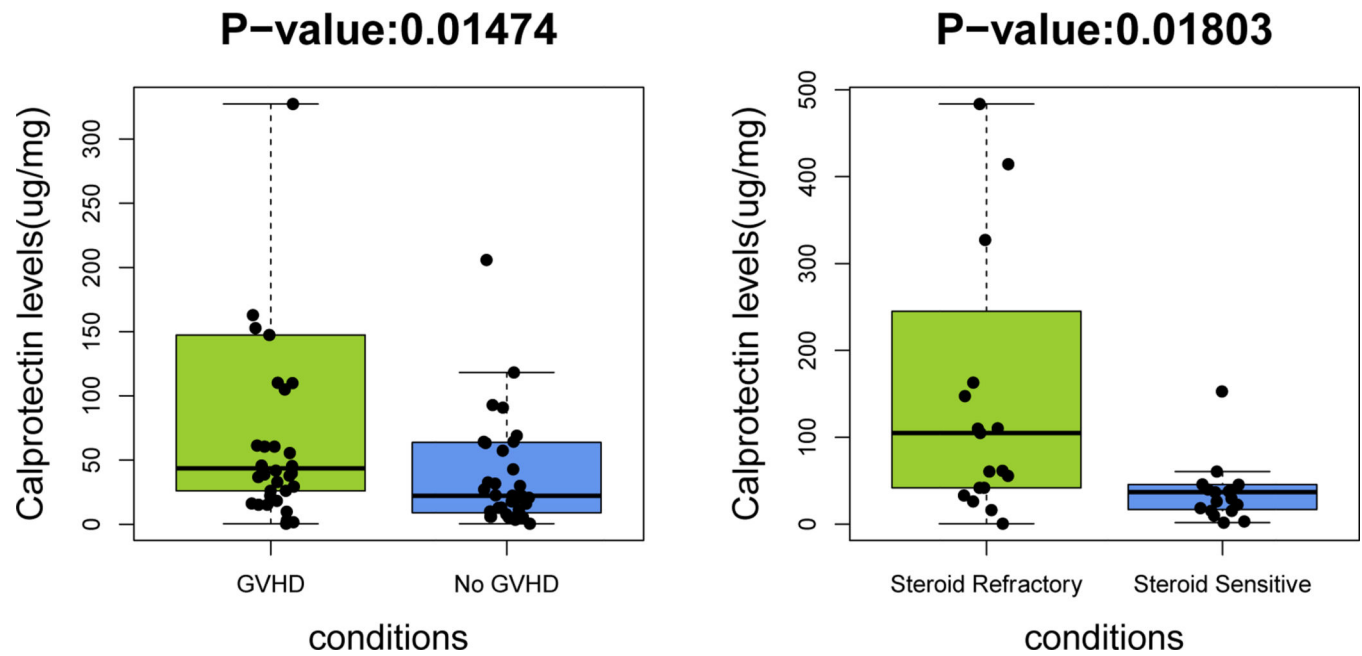


Figure 3. Fecal calprotectin levels ($\mu\text{g}/\text{mg}$) were significantly higher in patients with GVHD compared with patient without GVHD, and higher in patients with steroid-refractory GI GVHD compared with steroid-sensitive GI GVHD.

Fecal calprotectin levels ($\mu\text{g}/\text{mg}$) were measured using an ELISA-based method and compared using the Wilcoxon rank sum test. Outliers (8 in the GVHD vs. No GVHD plot, 4 in the Steroid Refractory vs. Steroid Sensitive plot) were included in the statistical models and excluded from bar plots to improve visualization.

Table 1

Category	No GVHD (N=38)	Steroid-Refractory GVHD (N=19)	Steroid-Sensitive GVHD (N=19)	Overall (N=76)
Age at BMT, Mean (SD)	50.6 (13.5)	53.6 (15.4)	53.6 (13.8)	52.1 (14.0)
Female sex, n (%)	19 (50.0%)	12 (63.2%)	6 (31.6%)	37 (48.7%)
Female donor, male recipient, n (%)	6 (15.8%)	3 (15.8%)	6 (31.6%)	15 (19.7%)
Underlying Disease, n (%)				
ALL	6 (15.8%)	5 (26.3%)	3 (15.8%)	14 (18.4%)
AML	14 (36.8%)	6 (31.6%)	4 (21.1%)	24 (31.6%)
CML	4 (10.5%)	0 (0%)	3 (15.8%)	7 (9.2%)
MDS/MF	7 (18.4%)	5 (26.3%)	7 (36.8%)	19 (25.0%)
NHL	4 (10.5%)	1 (5.3%)	1 (5.3%)	6 (7.9%)
Other	3 (7.9%)	2 (10.5%)	1 (5.3%)	6 (7.9%)
Conditioning Regimen, n (%)				
Myeloablative	14 (36.8%)	7 (36.8%)	10 (52.6%)	31 (40.8%)
Non-myeloablative	4 (10.5%)	0 (0%)	1 (5.3%)	5 (6.6%)
Reduced Intensity	20 (52.6%)	12 (63.2%)	8 (42.1%)	40 (52.6%)
Donor Type, n (%)				
MRD	13 (34.2%)	5 (26.3%)	3 (15.8%)	21 (27.6%)
UCB	5 (13.2%)	3 (15.8%)	2 (10.5%)	10 (13.2%)
URD	20 (52.6%)	11 (57.9%)	14 (73.7%)	45 (59.2%)
HLA Mismatch, n (%)				
0	31 (81.6%)	13 (68.4%)	13 (68.4%)	57 (75.0%)
1	2 (5.3%)	3 (15.8%)	4 (21.1%)	9 (11.8%)
Mismatched	5 (13.2%)	2 (10.5%)	2 (10.5%)	9 (11.8%)
Graft Source, n (%)				
Bone Marrow	6 (15.8%)	5 (26.3%)	5 (26.3%)	16 (21.1%)
Peripheral Blood	27 (71.1%)	11 (57.9%)	12 (63.2%)	50 (65.8%)
Umbilical Cord Blood	5 (13.2%)	3 (15.8%)	2 (10.5%)	10 (13.2%)
Days to Engraftment, Mean (SD)	21.6 (12.9)	17.7 (4.16)	18.8 (4.55)	19.9 (9.58)
GVHD Prophylaxis, n (%)				
CSP	1 (2.6%)	0 (0%)	0 (0%)	1 (1.3%)
CSP/MMF	6 (15.8%)	2 (10.5%)	4 (21.1%)	4 (5.3%)
CSP/MTX	2 (5.3%)	0 (0%)	0 (0%)	1 (1.3%)
FK/MMF	5 (13.2%)	2 (10.5%)	1 (5.3%)	8 (10.5%)
FK/MTX	23 (60.5%)	15 (78.9%)	14 (73.7%)	51 (67.1%)
SRL	1 (2.6%)	0 (0%)	0 (0%)	1 (1.3%)
Acute GVHD Grade, n (%)				
1	-	1 (5.3%)	0 (0%)	1 (1.3%)
2	-	2 (10.5%)	13 (68.4%)	15 (19.7%)
3	-	14 (73.7%)	3 (15.8%)	17 (22.4%)
4	-	2 (10.5%)	1 (5.3%)	3 (3.9%)
GI GVHD Stage, n (%)	-			

Category	No GVHD (N=38)	Steroid-Refractory GVHD (N=19)	Steroid-Sensitive GVHD (N=19)	Overall (N=76)
1	-	3 (15.8%)	14 (73.7%)	17 (22.4%)
2	-	2 (10.5%)	1 (5.3%)	3 (3.9%)
3	-	3 (15.8%)	3 (15.8%)	6 (7.9%)
4	-	11 (57.9%)	1 (5.3%)	12 (15.8%)
Skin GVHD, n (%)	-	6 (31.6%)	7 (36.8%)	13 (17.1%)
Liver GVHD, n (%)	-	5 (26.3%)	4 (21.1%)	9 (11.8%)
1-year Overall Survival, n (%)	30 (78.9%)	5 (26.3%)	15 (78.9%)	50 (65.8%)
Fluoroquinolone	30 (78.9%)	17 (89.5%)	16 (84.2%)	63 (82.9%)
Cephalosporin	20 (52.6%)	11 (57.9%)	10 (52.6%)	41 (53.9%)
Carbapenem	11 (28.9%)	9 (47.4%)	6 (31.6%)	26 (34.2%)
Aminoglycoside	5 (13.2%)	2 (10.5%)	0 (0%)	7 (9.2%)
IV Vancomycin	20 (52.6%)	9 (47.4%)	10 (52.6%)	39 (51.3%)
PO Vancomycin	3 (7.9%)	1 (5.3%)	2 (10.5%)	6 (7.9%)
Other Antibiotic	11 (28.9%)	6 (31.6%)	8 (42.1%)	25 (32.9%)
1-Year BSI, n (%)	9 (23.7%)	9 (47.4%)	4 (21.1%)	22 (28.9%)
1-Year <i>C. difficile</i> infection, n (%)	3 (7.9%)	2 (10.5%)	5 (26.3%)	10 (13.2%)